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Realization of Baseband Signal Processing for Beidou/GPS Multi-Mode Receiver by FPGA

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***Abstract*—Beidou and GPS are an important part of the GNSS (Global Navigation Satellite System). Beidou is a global satellite navigation system that is independent development and operation in China. GPS is the most mature development system in the world currently. The research on the Beidou/GPS multi-mode receiver has an important practical significance at present. Compared with hardware receiver, software receiver realizes the performance of the receiver by software, and is suitable for developing new algorithms. Therefore it has advantages of flexibility, convenience, and low costs. A method to realize the baseband signal processing of the Beidou/GPS multi-mode receiver is introduced in this paper by using the chip EP2S90F1020I4, which is Stratix II series of Altera's FPGA. Using Verilog HDL to design the baseband processing module of Beidou/GPS, including the fast acquisition module and tracking channel processing module, and the tracking channel processing module includes the carrier/code NCO generation module, the complex phase rotation DDC (Digital Down Converter) module, the Beidou/GPS pseudo-random code generator, code correlator module and frame synchronization module, etc. Test results show that the baseband signal processing module based on FPGA with the DSP calculation module can achieve the normal PVT calculation.**

Keywords-Beidou; GPS; FPGA; Verilog; PVT

I. INTRODUCTION

Beidou satellite navigation system is one of the GNSS, which independently developed by China, in order to provide high quality positioning, navigation and timing (PNT) services for global users, and also has the function of short message communication [1-2]. GPS is the most mature development, and has the most users currently. It has practical significance to research Beidou/GPS multi-mode receiver at present.

The traditional hardware receiver has the advantages of high sensitivity and high signal processing speed, but it also has the disadvantages, such as poor flexibility and large size. As a result, it can't update the structure after it is made. With the progress of technology, the functions of baseband signal processing can be achieved by software. So it can meet the requirement of flexibility in the receiver. It is also convenient to upgrade, suitable for developing new algorithm and improving the performance of the receiver. FPGA (large scale Field Programmable Gate Array) have rich resources of digital logic and many excellent development kits, such as ALTERA's Quartus II, XILINX's Vivado and so on. Therefore the FPGA are suitable for being a carrier of the receiver. In addition, if we want to develop deeply-coupled and integrated navigation system, we should do some major changes of internal structure in the receiver. Therefore, the software receiver has the advantage of re-configurability, and the traditional hardware receiver is difficult to realize [3-5].

The receiver has the functions of RF (Radio Frequency) front-end processing, baseband signal processing, positioning and correlation calculating, and so on. Baseband signal processor mainly deals with the IF (Intermediate Frequency) signal, after the RF front-end processor, which should contain information such as location, speed, time (PVT), and so on. In the paper, we use FPGA greatly to achieve the functions of baseband signal processing, and the test results show that it can normally realize the functions of baseband signal processing, and the ability of searching satellite is good.

II. DESIGN OF BASEBAND SIGNAL PROCESSOR BASED ON FPGA

A. Block Diagram of Baseband Signal Processing

The receiver we designed needs to receive BD2 B1 and GPS L1 dual-mode signal. The frequency of BD2 B1 RF signal is 1561.098MHz, and GPS L1 is 1575.42MHz. Receiving antenna receives the dual-mode signal, after the low noise amplifier, through the center frequency of 1568 MHz bandwidth 40 MHz filter filtering the out-of-band interference signal, with 1568 MHz local oscillation signal after AD8347 chips for mixing, get two I/Q orthogonal IF baseband signals, the two orthogonal signals access FPGA processing baseband signal after the chip of AD9233. The overall block diagram of multi-mode receiver baseband signal processor is shown in Figure 1.

It can be seen from Figure 1 that the two I/Q orthogonal baseband signals are removed the IF carrier through the first level DDC, and get close to zero intermediate frequency of baseband signal. Moreover the satellite is moving relative to the receiver in the ground, and this leads to the frequency received and the frequency from satellite signal is different. Thus the Doppler shift of the signal is generated. Therefore,

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the baseband signal received through the first level DDC are close to zero intermediate frequency and it is required to carry out the secondary DDC in order to filter out the rest of the Doppler frequency. The signal get rid of the Doppler frequency shift performs the pseudo-code correlation operations to remove the pseudo-code signal. The rest of the signals are the navigation message. Stripping of the carrier and pseudo-random code is completed by carrier NCO and code NCO. DSP tests bit synchronization and frame synchronization state to initialize the bit and frame synchronization count value in the frame synchronization processing module, combined with the code phase value produced by code NCO and carrier phase value of carrier NCO to generate pseudo-range information for DSP to read [6-7].

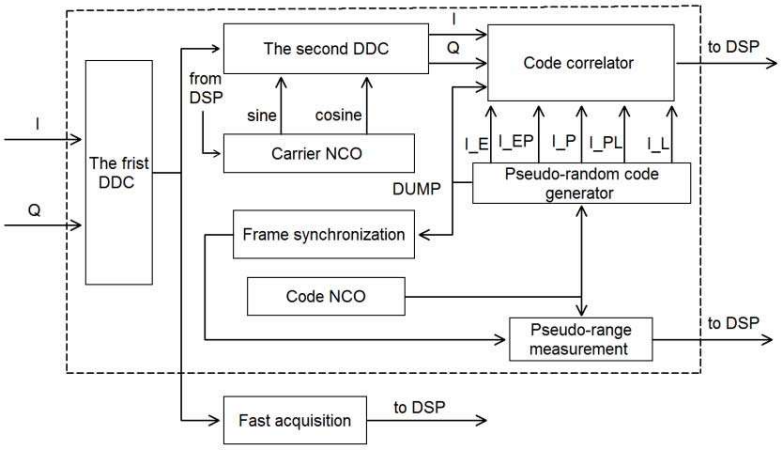

Figure1. Block diagram of Beidou/GPS baseband signal processing

In baseband signal processing module, we use the FPGA, which is pure hardware circuit, has the characteristics of high speed parallel. We use Verilog HDL language to design the circuits, make full use of rich internal resources in the chip EP2S90F1020I4, which is Stratix II series of Altera's FPGA. Consequently one can better realize the functions of baseband signal processing. The design has the advantages that the hardware can be upgrade online.

B. Main Module Design by FPGA

1) Digital Down Converter (DDC)

The design is divided into two levels for a better implementation of digital down conversion. The phase rotation method is employed to realize digital down conversion. The first level DDC remove the IF carrier component respectively in Beidou and GPS signals. The second level DDC strip for Doppler frequency shift after capturing the satellite signal and entering the tracking channel.

DDC module consists of carrier NCO module and the complex phase rotation module .The structure is shown in Figure 2.

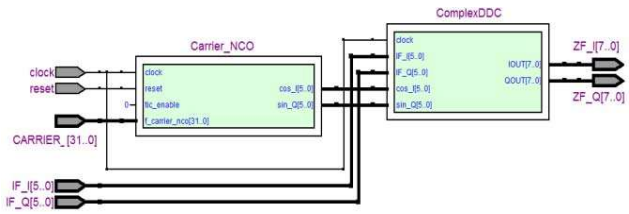

Figure 2. Block diagram of DDC module

The frequency control word of Carrier NCO module in first level DDC is corresponding to the center frequency of the IF. For BD2 B1 signal, because its frequency is 1561.098 MHz, the center frequency of RF filter is 1568 MHz, so the frequency value corresponding to the frequency control word value is -6.902MHz. For GPS L1 signal, because its frequency is 1575.42MHz, so the frequency value corresponding to the frequency control words is 7.42MHz. The cosine function value produced by Carrier NCO module and AD collection of *I/Q* two orthogonal frequency baseband signals as the input of ComplexDDC module, output of ComplexDDC module system get zero intermediate frequency signal whose intermediate frequency carrier component is removed. ComplexDDC module is used to make a whole movement on the entire spectrum. So the collected signals in the front can be moved to near zero frequency.

The rate of IF signals, which is collected and quantified by AD chips in the RF front-end circuit, is 62MSPS (Sample per Second). The frequency of carrier $I + Q$ j is ω_c . The carrier NCO generates the complex signal with the same data rate is $\cos(\omega_n \cdot t) + j \cdot \sin(\omega_n \cdot t)$, both do complex multiplication as follow.

$$(I + Q \cdot j) \times (\cos(\omega_n \cdot t) + j \cdot \sin(\omega_n \cdot t)) = (I\cos(\omega_n \cdot t) - Q\sin(\omega_n \cdot t)) + j \cdot (I\sin(\omega_n \cdot t) + Q\cos(\omega_n \cdot t)) \quad (1)$$

By Eq. (1), with the plural multiplication results, we set $I\cos(\omega_n \cdot t) - Q\sin(\omega_n \cdot t)$ as a new branch *I*, $I\sin(\omega_n \cdot t) + Q\cos(\omega_n \cdot t)$ as a new branch *Q*. And they are

used for correlating and accumulating calculation [8]. Use the above phase rotation method to achieve the frequency down conversion. We can eliminate the intermediate frequency carrier when $\omega_n = -\omega_c$ by Eq. (1).

The frequency is close to zero after the first DDC, because of the relative motion of the receiver and the satellite, the received signal exist Doppler frequency offset and the local clock error, etc. In order to eliminate the deviation, we can make the signal through the second DDC module to get zero frequency baseband signals. Compared with the traditional orthogonal frequency down conversion method, the method of phase rotation can realize digital down conversion without low-pass filter, and reduce delay and attenuation of the signal. In the second DDC, the frequency control word of Carrier NCO module is provided by DSP. DSP read the output of satellite fast acquisition module, calculate the results and then get the frequency control word needed.

2) Carrier NCO and code NCO module

The function of Carrier NCO module is to produce single complex sine and cosine signal whose frequency can be adjusted, and make them as a input signal on the back end of the complex DDC module, used for removing the IF carrier components and strip carrier for Doppler frequency shift in every track channel [8]. The frequency control word of sine and cosine function value generated by carrier NCO module is corresponding to the center frequency of intermediate frequency signal, calculated and provided by DSP. Figure 3 depicts the schematic diagram of Carrier NCO.

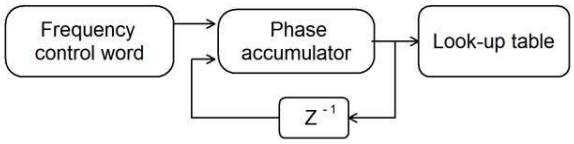

Figure 3. Schematic diagram of carrier NCO

Look-up table method (LUT) is a fast and easy way to realize the Carrier NCO. That is according to every sine or cosine signal phase to calculate the values in advance, and uses the phase angle as an address to store the corresponding sine or cosine data.

When the frequency control word is received by Carrier NCO module, it is input into the phase accumulator. The accumulator drives the clock to start counting. When the counted value is equal to the value of frequency control word, it is input the accumulated value to the phase adder, combined with the initial phase. And we obtain the sine and cosine values through the look-up table. The bit width of the frequency control word and phase of the accumulator of the Carrier NCO module are 32-bit, the address width of ROM memory and the width of the sine and cosine signal data are 6-bit.

The structure of Code NCO module is similar to that of Carrier NCO, but it is the overflow of the code phase accumulation as the output. It is used as a clock signal of pseudo-random code generator. Thus the structure is simpler. The block diagram of the Code NCO module is given in Figure 4.

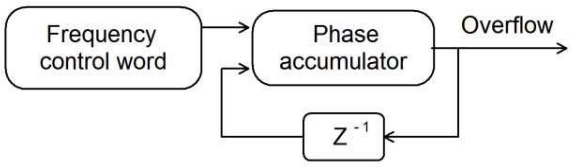

Figure 4. Block diagram of code NCO

The frequency control word and phase accumulator of the Code NCO module we designed are 32-bit, and the output is the overflow bit of phase accumulator, that is the thirty-third bit of phase accumulator. The working clock of phase accumulator is 62MHz, which can output 4×2.046 MHz and 4×1.023 MHz clock frequency according to the processing channels of Beidou and GPS signals respectively.

3) Pseudo-random code generator

The *C* code rate of Beidou is 2.046MCPS (Chips per Second), 2046 yards long [9]. The *C/A* code rate of GPS is 1.023Mcps, 1023 yards long. They are generally through the way of linear shift register to produce pseudo-random code in real time. In order to be more efficient, convenient, and economical to produce pseudo-random code, we adopt a method that solidified pseudo-random code sequence of satellite in the dual-port RAM of FPGA. In the process of tracking according to the corresponding satellite number to read data for pseudo-random code in the dual port RAM. As for Beidou satellite, writing the *C* code MIF data firmed in the dual port RAM is written as follows.

```
WIDTH=37;
DEPTH=2046;
ADDRESS_RADIX=UNS;
DATA_RADIX=BIN;
CONTENT BEGIN
0 : 1011011010010111010010101101101011010;
1 : 0100100101101000101101010010010100101;
2 : 1011011010010111010010101101110100101;
3 : 0100100101101000101101010010101011010;
4 : 1011011010010111010011010010010100100;
5 : 0100100101101001010010101100101011001;
   ...........
```

The width of the data is 37, corresponding to Beidou satellite number. The depth of the data is 2046, corresponding to the C code length of Beidou satellites. Code value from different satellites in the same position consists of a MIF file data. We realize the function of pseudo-random code generation using the Verilog, produce five paths pseudo-random code delay code sequence through four non-blocking assignment statements in the process of Verilog. The design of Beidou/GPS pseudo-random code generator in this way can meet the design requirements more efficiently.

4) Code correlator module

One of the code correlator module inputs is *I/Q* complex baseband signal after the module of second DDC. The other input are the five ways pseudo-random delay code sequence which produced by pseudo-random code generator and delay 1/4 chip respectively, including early code, early prompt code, prompt code, prompt lag code and lag code. The five

pseudo-random code sequences do correlation accumulation operations respectively to the code with complex baseband signal after the second DDC. The result is send to the DSP module in order to keep a continuous tracking of the satellite signal with fine adjustment Code NCO and Carrier NCO parameters. The formula is described as Eq. (2).

$$F(m)=\sum_{n=0}^{N} R(n+\mathrm{m})\cdot G(n) \tag{2}$$

F (m) is correlation accumulation value of a signal in the period of accumulation, R (n) is satellite signal received by receiver, the width is 8 bits in this design, the data rate is 62 MHz, G (n) is the pseudo-random code which produced by local, the value of G (n) is ± 1. Therefore we can transform the operation of multiplication in Eq. (2) to the operation of addition and subtraction, using G (n) to control the accumulation results of R (n + m).

The key code of Verilog to realize the correlation operation is given as follows.

```
always @(posedge clock)
begin
        if(reset==1'b1)  result <= 0;
           else  if(sload==1'b1)
                        result <= {{14{data[7]}},data};
            else
              begin
               if(add_sub==1'b1)
                  result <= result + {{14{data[7]}},data};
               else
                   result <= result - {{14{data[7]}},data};
             end
 end
```

The "*data*" in the program are the input signal of baseband *I* or *Q*, "*add_sub*" denotes the input signal *G(n)*, "*sload*" marks the control signal of correlative accumulative cycle which indicate the beginning of pseudo-random code cycle sequence with the rising edge, "*result*" is the accumulation output of correlation code, the time of calculation is 1ms, the width of accumulative value is 22bits.

5) Frame synchronization processing module

Beidou/GPS satellite navigation messages are sent to the terminal in the form of binary. The data after demodulation is still not correct if without the frame synchronization detection. So frame synchronization is needed to detect the start and the end of each frame position, and then decode for the data outside each frame header, in order to get the broadcast information.

When the signals get into the stable tracking, detect the state of bit and frame synchronization state by DSP module, if the frame synchronization has been sure, and then initialize the count of frame synchronization module and bit synchronization module, the former has the precision of 20 ms and the latter is 1ms. Through the code phase value of Code NCO and carrier phase value of Carrier NCO calculate the pseudo-range, which provide the information for DSP module to calculate.

III. HARDWARE TEST OF BASEBAND SIGNAL PROCESSOR

Verilog HDL language is used to design and simulate each module of channel baseband signal processor. Download them to the chip EP2S90F1020I4 which is the StratixII series FPGA of ALTERA. When all expected functions have been finished, combined with the DSP module of TI to realize the functions of position calculating and loop tracking. The hardware test results are shown in Figure 5.

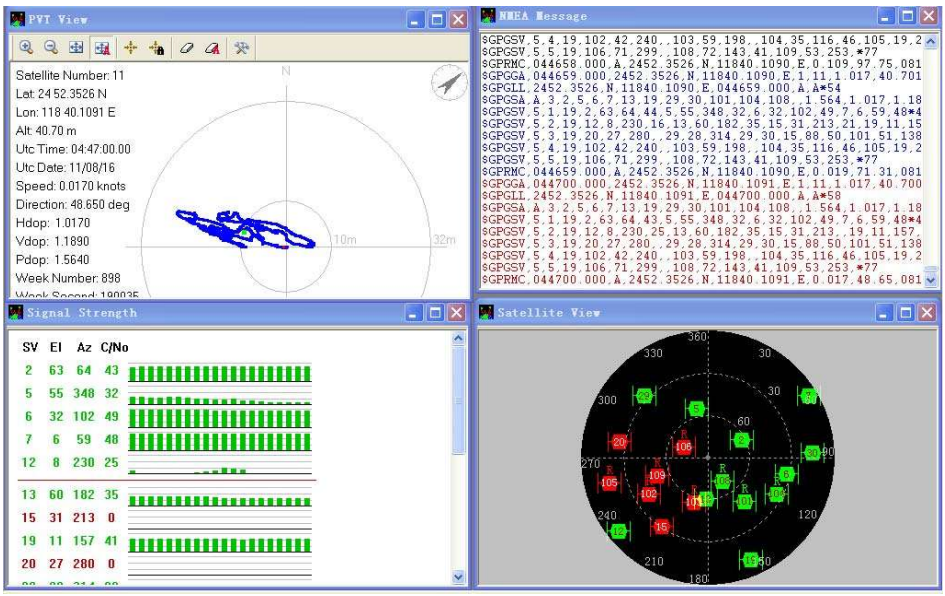

Figure 5. Hardware testing of BDS/GPS receiver

From Figure 5 one can find that the 11 satellites are searched at present, including 8 GPS satellites, 3 Beidou satellites. Receiver positioning results is shown on the upper left part of Fig.5, and the blue block shows the positioning trajectory of receiver, one point per second, green block is the location of the average. The upper right part of the Fig.5 displays NMEA-0183 data that receiver decoded. The lower left part of Fig.5 shows carrier-to-noise density ratio of searched satellites. Fig.5 lower right part is the satellite sky chart, which can see the number of satellite searched by receiver and its distribution. In this chart, the green satellites represents that they have been received and participated in calculating , red represents that they should have been predicted, but by reason of building block could not receive the satellite signal.

Hardware test results demonstrate that the baseband signal processor designed by FPGA, can realize the prescribed function.

IV. CONCLUSION

Using Verilog HDL language to implement the function of Beidou/GPS baseband signal tracking channel process on FPGA, in combination with DSP decoding module, can achieve the function of receiver positioning and navigation. This design program structure is re-configurability, compact, modular and easy to upgrade. It is also suitable for developing a multi-mode receiver, which has great practical significance.

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